18.100C Lecture 4 Summary

Definition of metric space. Various example (including French railroad metric, Hamming distance).

Theorem 4.1. *(Triangle inequality for the Euclidean norm) On $\mathbb{R}^n$, define $\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$. Then $\|x + y\| \leq \|x\| + \|y\|$.*

For general metric spaces: ball neighbourhoods. Open subsets.

Theorem 4.2. *Every ball neighbourhood is an open subset.*

MIT OpenCourseWare
http://ocw.mit.edu

18.100C Real Analysis

Fall 2012

For information about citing these materials or our Terms of Use, visit: http://ocw.mit.edu/terms.